Problem Statement

Evaluate the following limit:

$$\lim_{n\to\infty} \frac{1}{n^3} \sum_{k=1}^{n-1} \frac{\sin\left(\frac{(2k-1)\pi}{2n}\right)}{\cos^2\left(\frac{(k-1)\pi}{2n}\right)\cos^2\left(\frac{k\pi}{2n}\right)}$$

Detailed Solution

This limit can be elegantly evaluated by transforming the summand into a telescoping series using trigonometric identities. Let's break this down step-by-step.

Step 1: Establish a useful trigonometric identity

Let's analyze the expression $\tan^2 A - \tan^2 B$. Converting to sine and cosine, we get:

$$\tan^2 A - \tan^2 B = \frac{\sin^2 A}{\cos^2 A} - \frac{\sin^2 B}{\cos^2 B}$$

Finding a common denominator gives:

$$\tan^2 A - \tan^2 B = \frac{\sin^2 A \cos^2 B - \cos^2 A \sin^2 B}{\cos^2 A \cos^2 B}$$

Now, we can manipulate the numerator using the Pythagorean identity $\cos^2\theta = 1 - \sin^2\theta$:

$$\begin{aligned}
\sin^2 A \cos^2 B - \cos^2 A \sin^2 B &= \sin^2 A(1 - \sin^2 B) - (1 - \sin^2 A)\sin^2 B \\
&= \sin^2 A - \sin^2 A \sin^2 B - \sin^2 B + \sin^2 A \sin^2 B \\
&= \sin^2 A - \sin^2 B
\end{aligned}$$

Next, we factor this difference of squares and apply the sum-to-product identities:

$$\begin{aligned}
\sin^2 A - \sin^2 B &= (\sin A - \sin B)(\sin A + \sin B) \\
&= \sin(A - B)\sin(A + B)
\end{aligned}$$

Substituting this back into our original fraction, we obtain our master identity:

$$\tan^2 A - \tan^2 B = \frac{\sin(A - B)\sin(A + B)}{\cos^2 A \cos^2 B}$$

Step 2: Apply the identity to the summand

Let's look at the general term of the sum in our limit. Let:

- $A = \frac{k\pi}{2n}$
- $B = \frac{(k-1)\pi}{2n}$

Notice what happens when we add and subtract A and B:

$$A + B = \frac{k\pi}{2n} + \frac{(k-1)\pi}{2n} = \frac{(2k-1)\pi}{2n}$$
$$A - B = \frac{k\pi}{2n} - \frac{(k-1)\pi}{2n} = \frac{\pi}{2n}$$

Observe that $A+B$ perfectly matches the argument of the sine function in the numerator of our problem, and the arguments of the cosine functions in the denominator match B and A perfectly.

Using our identity from Step 1, we can write:

$$\tan^2\left(\frac{k\pi}{2n}\right) - \tan^2\left(\frac{(k-1)\pi}{2n}\right) = \frac{\sin\left(\frac{\pi}{2n}\right)\sin\left(\frac{(2k-1)\pi}{2n}\right)}{\cos^2\left(\frac{k\pi}{2n}\right)\cos^2\left(\frac{(k-1)\pi}{2n}\right)}$$

Isolating the fraction that appears in our problem, we get:

$$\frac{\sin\left(\frac{(2k-1)\pi}{2n}\right)}{\cos^2\left(\frac{k\pi}{2n}\right)\cos^2\left(\frac{(k-1)\pi}{2n}\right)} = \frac{1}{\sin\left(\frac{\pi}{2n}\right)}\left[\tan^2\left(\frac{k\pi}{2n}\right) - \tan^2\left(\frac{(k-1)\pi}{2n}\right)\right]$$

Step 3: Evaluate the telescoping sum

We can now rewrite the summation portion of the limit, pulling the constant factor $\frac{1}{\sin\left(\frac{\pi}{2n}\right)}$ out of the sum since it does not depend on the index k:

$$\sum_{k=1}^{n-1}\left[\frac{1}{\sin\left(\frac{\pi}{2n}\right)}\left(\tan^2\left(\frac{k\pi}{2n}\right) - \tan^2\left(\frac{(k-1)\pi}{2n}\right)\right)\right]$$
$$= \frac{1}{\sin\left(\frac{\pi}{2n}\right)}\sum_{k=1}^{n-1}\left[\tan^2\left(\frac{k\pi}{2n}\right) - \tan^2\left(\frac{(k-1)\pi}{2n}\right)\right]$$

Because this is a telescoping series, all intermediate terms cancel out. We are left with only the final positive term (when $k = n-1$) and the initial negative term (when $k = 1$):

$$= \frac{1}{\sin\left(\frac{\pi}{2n}\right)}\left[\tan^2\left(\frac{(n-1)\pi}{2n}\right) - \tan^2(0)\right]$$

Since $\tan(0) = 0$, the second term vanishes. For the first term, we can simplify the argument:

$$\tan^2\left(\frac{(n-1)\pi}{2n}\right) = \tan^2\left(\frac{\pi}{2} - \frac{\pi}{2n}\right)$$

Using the co-function identity $\tan\left(\frac{\pi}{2} - \theta\right) = \cot(\theta)$, this becomes:

$$= \frac{1}{\sin\left(\frac{\pi}{2n}\right)} \cot^2\left(\frac{\pi}{2n}\right) = \frac{\cos^2\left(\frac{\pi}{2n}\right)}{\sin^3\left(\frac{\pi}{2n}\right)}$$

Step 4: Compute the final limit

Now, substitute the simplified sum back into the original limit:

$$L = \lim_{n\to\infty} \frac{1}{n^3} \left[\frac{\cos^2\left(\frac{\pi}{2n}\right)}{\sin^3\left(\frac{\pi}{2n}\right)}\right]$$

To make this easier to evaluate, let $x = \frac{\pi}{2n}$. As $n \to \infty$, it follows that $x \to 0$. Rearranging for n, we get $n = \frac{\pi}{2x}$, which means $n^3 = \frac{\pi^3}{8x^3}$. Substituting x into our limit gives:

$$L = \lim_{x\to 0} \left(\frac{8x^3}{\pi^3}\right) \frac{\cos^2 x}{\sin^3 x}$$

Grouping the terms to utilize standard limit properties:

$$L = \frac{8}{\pi^3} \lim_{x\to 0} \left[\left(\frac{x}{\sin x}\right)^3 \cos^2 x\right]$$

Since we know the standard limits $\lim_{x\to 0} \frac{x}{\sin x} = 1$ and $\lim_{x\to 0} \cos x = 1$, we can evaluate directly:

$$L = \frac{8}{\pi^3} \cdot (1)^3 \cdot (1)^2 = \frac{8}{\pi^3}$$

Final Answer:

$$\boldsymbol{\frac{8}{\pi^3}}$$